SDU
University of Southern Denmark

Number of Subgraphs and Their Converses in Tournaments and New Digraph Polynomials

Ai, Jiangdong; Gutin, Gregory; Lei, Hui; Yeo, Anders; Zhou, Yacong

Published in:
Journal of Graph Theory

DOI:
10.1002/jgt.23257

Publication date:
2025

Document version:
Final published version

Document license:
CC BY

Citation for pulished version (APA):
Ai, J., Gutin, G., Lei, H., Yeo, A., & Zhou, Y. (2025). Number of Subgraphs and Their Converses in Tournaments and New Digraph Polynomials. *Journal of Graph Theory*, *110*(2), 127-131. https://doi.org/10.1002/jgt.23257

Go to publication entry in University of Southern Denmark's Research Portal

Terms of use
This work is brought to you by the University of Southern Denmark.
Unless otherwise specified it has been shared according to the terms for self-archiving.
If no other license is stated, these terms apply:

- You may download this work for personal use only.
- You may not further distribute the material or use it for any profit-making activity or commercial gain
- You may freely distribute the URL identifying this open access version

If you believe that this document breaches copyright please contact us providing details and we will investigate your claim.
Please direct all enquiries to puresupport@bib.sdu.dk

Download date: 29. Oct. 2026

Check for updates

Journal of Graph Theory

WILEY

ARTICLE OPEN ACCESS

Number of Subgraphs and Their Converses in Tournaments and New Digraph Polynomials

Jiangdong Ai[1] | Gregory Gutin[1,2] | Hui Lei[3] | Anders Yeo[4,5] | Yacong Zhou[2]

[1]School of Mathematical Sciences and LPMC, Nankai University, Tianjin, China | [2]Department of Computer Science, Royal Holloway University of London, Surrey, UK | [3]School of Statistics and Data Science, LPMC and KLMDASR, Nankai University, Tianjin, China | [4]Department of Mathematics and Computer Science, University of Southern Denmark, Odense, Denmark | [5]Department of Mathematics and Applied Mathematics, University of Johannesburg, Auckland Park, South Africa

Correspondence: Gregory Gutin (g.gutin@rhul.ac.uk)

Received: 23 July 2024 | **Revised:** 5 February 2025 | **Accepted:** 9 April 2025

Funding: This study was supported by National Natural Science Foundation of China under grant (Nos. 12161141006, 12401456), Partially supported by the NSFC grant (No. 12371351).

Keywords: converse | digraph polynomial | number of subgraphs | oriented graph | tournaments

ABSTRACT

An oriented graph D is *converse invariant* if, for any tournament T, the number of copies of D in T is equal to that of its converse $-D$. El Sahili and Ghazo Hanna [J. Graph Theory 102 (2023), 684-701] showed that any oriented graph D with maximum degree at most 2 is converse invariant. They proposed a question: Can we characterize all converse invariant oriented graphs? In this paper, we introduce a digraph polynomial and employ it to give a necessary condition for an oriented graph to be converse invariant. This polynomial serves as a cornerstone in proving all the results presented in this paper. In particular, we characterize all orientations of trees with diameter at most 3 that are converse invariant. We also show that all orientations of regular graphs are not converse invariant if D and $-D$ have different degree sequences. In addition, in contrast to the findings of El Sahili and Ghazo Hanna, we prove that every connected graph G with maximum degree at least 3, admits an orientation D of G such that D is not converse invariant. We pose a new conjecture.

1 | Introduction

We refer to [1] for terminology and notation not introduced here. All graphs considered throughout this paper contain neither loops nor multiple edges. An *arc* in a digraph from vertex u to vertex v is denoted by (u, v). An *orientation* of a graph G is a digraph D obtained from G by replacing every edge uv by either arc (u, v) or arc (v, u). A digraph with at most one arc between any pair of vertices is called an *oriented graph* (or *orgraph* for short). A *tournament* is an orientation of a complete graph. A *Hamiltonian dipath* in a tournament T is a directed path containing all vertices of T. Hamiltonian dicycles are defined similarly. Let $D = (V(D), A(D))$ be a digraph. For a vertex $v \in V(D)$, we denote by $d_D^+(v)$ and $d_D^-(v)$ the out-degree and in-degree of v, respectively. The *degree* of v in D, denoted by $d_D(v)$, is the sum of its out-degree and in-degree in D. We may drop the D subscripts if D is clear from the context. We use $\Delta(H)$ to denote the maximum degree of the (di)graph H.

Counting subdigraphs in a tournament is a well-known problem. The lower and upper bounds for the maximum number $P(n)$ of Hamiltonian dipaths in a tournament on n vertices were intensively studied. Szele [2] showed that $\frac{n!}{2^{n-1}} \le P(n) \le O\left(\frac{n!}{2^{3n/4}}\right)$. Alon [3] improved the upper bound by showing that $P(n) \le O\left(n^{3/2}\frac{n!}{2^{n-1}}\right)$. Adler, Alon and Ross [4] improved the lower bound to $(e - o(1))\frac{n!}{2^{n-1}}$ and asked whether $P(n) = \Theta\left(\frac{n!}{2^{n-1}}\right)$. Fridgut and Kahn [5] improved the upper

This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

© 2025 The Author(s). *Journal of Graph Theory* published by Wiley Periodicals LLC.

bound for $P(n)$ to $O\left(n^{3/2-\xi}\frac{n!}{2^{n-1}}\right)$, where $\xi \in [0.2507, 0.2508]$. Moon [6] examined the upper and lower bounds for the minimum number of Hamiltonian dipaths in strong tournaments on n vertices. Busch [7] determined the minimum number of Hamiltonian dipaths in strong tournaments on n vertices. Relevant results about Hamiltonian dicycles can be found in [3–5, 8–11]. We refer the readers to [12] for the problem of counting more general spanning subdigraphs, such as orientations of regular graphs, in tournaments.

In this paper, we compare the number of subdigraphs and their converses in every tournament T. The *converse* of D, denoted by $-D$, is the digraph on the same vertex set where $(u, v) \in A(-D)$ if and only if $(v, u) \in A(D)$. Let $f_T(D)$ denote the number of copies of D in a tournament T. An ograph D is *converse invariant* if $f_T(D) = f_T(-D)$ for every tournament T.

An oriented path $P = x_1x_2...x_p$ is *antidirected* if for every $p - 1 \geq i \geq 2$, $x_ix_{i+1} \in A(P)$ if and only if $x_ix_{i-1} \in A(P)$. Rosenfeld [13] proved that antidirected paths are converse invariant. El Sahili and Ghazo Hanna [14] generalized Rosenfeld's result for any type of oriented paths, and also for cycles. They showed the following result.

Theorem 1.1 [14]. *Let G be a graph with $\Delta(G) \leq 2$. Then, every orientation D of G is converse invariant.*

In the same paper, El Sahili and Ghazo Hanna proposed the following natural problem.

Problem 1.2 [14]. Can we characterize all converse invariant ographs?

In this paper, we introduce the following digraph polynomial of x

$$P_D(x) = \sum_{u \in V(D)} (1 + x)^{d_D^+(u)}(1 - x)^{d_D^-(u)},$$

and use it to give the following necessary condition for an ograph to be converse invariant.

Theorem 1.3. *Let D be a converse invariant ograph. Then $P_D(x) = P_{-D}(x)$.*

Theorem 1.3 is utilized as a foundational tool to prove all results in this paper. In particular, we use Theorem 1.3 to show that paths and cycles are the only connected graphs with the property that, every orientation of them is converse invariant.

Theorem 1.4. *Let G be a connected graph with $\Delta(G) \geq 3$. Then, there exists an orientation D of G such that D is not converse invariant.*

Given a star $K_{1,d}$ with the central vertex v, we use $\overrightarrow{K}^i_{1,d}$ to denote the orientation of $K_{1,d}$ with $d^+(v) = i$. In [14], El Sahili and Ghazo Hanna showed that when $d \geq 3$, $\overrightarrow{K}^d_{1,d}$ is not converse invariant by constructing a special tournament. Building upon their work, our paper establishes, using our method, that all orientations D of stars $K_{1,d}$ with $d \geq 3$ are not converse invariant unless D is isomorphic to $-D$ (denoted by $D \cong -D$). Furthermore, we extend this by showing the following stronger result. Note that the edge-mirroring operation is introduced in [15]. We define it in Section 3.

Theorem 1.5. *Let D be an orientation of a tree with diameter at most three that is not a path. Then D is converse invariant if and only if $D \cong -D$, or D is formed by bridge-mirroring $\overrightarrow{K}^0_{1,2}$ or $\overrightarrow{K}^2_{1,2}$ at the central vertex.*

This paper is organized as follows. In Section 2, we begin by proving Theorem 1.3. Then using Theorem 1.3, we show that all orientations of regular graphs are not converse invariant if D and $-D$ have different degree sequences. At the end of this section, we prove Theorem 1.4. In Section 3, we apply our method to characterize all orientations of trees with diameter at most 3 that are converse invariant. An open problem is proposed in the subsequent section.

1.1 | Additional Notation

Given a digraph $D = (V(D), A(D))$. We use $|D|$ to denote the number of vertices and $\|D\|$ the number of arcs. We say that a vertex v is a *source* if $d_D^-(v) = 0$ and a *sink* if $d_D^+(v) = 0$. The *degree sequence* of D, denoted by $\mathrm{Deg}(D)$, is the multiset $\{(d_D^+(v), d_D^-(v)) : v \in V(D)\}$. Let $\mathrm{Deg}^+(D)$ and $\mathrm{Deg}^-(D)$ be the multisets $\{d_D^+(v) : v \in V(D)\}$ and $\{d_D^-(v) : v \in V(D)\}$, respectively. The digraph obtained by adding an arc (u, v) to D is denoted by $D + (u, v)$. For any pair of digraphs D and D', we use $\mathrm{ism}(D, D')$ to denote the number of isomorphisms from D to subdigraphs of D', and $\mathrm{ism}_{u\to v}(D, D')$ to denote the number of such isomorphisms given that $u \in V(D)$ is mapped to $v \in V(D')$. Let $\mathrm{aut}(D)$ denote the number of automorphisms of D. A tournament T is *transitive* if $(u, v) \in A(T)$ and $(v, w) \in A(T)$ imply that $(u, w) \in A(T)$. $P = x_1x_2...x_n$ is an (x_1, x_n)-*dipath* in D if $x_ix_{i+1} \in A(D)$ for all $1 \leq i < n$. A digraph D is *strong* if there exists a (u, v)-dipath for every ordered pair (u, v) of vertices in D. Let G be a graph. The *distance* between two vertices is defined as the length of the shortest path connecting them. The *diameter* of G is the greatest distance between any pair of vertices in G.

2 | A Polynomial of Degree Sequences

We start this section by proving Theorem 1.3.

Theorem 1.3. *Let D be a converse invariant ograph. Then $P_D(x) = P_{-D}(x)$.*

Proof. Let T be a random tournament on $|D| - 1$ vertices where each direction of the arc between any pair of vertices is taken independently and uniformly with probability 1/2. Let p be an arbitrary real in the interval $[-1/2, 1/2]$. Obtain a new random tournament from T by adding a new vertex v^*, and for any vertex $v \in V(T)$, we add the arc v^*v with probability $1/2 + p$ and vv^* with probability $1/2 - p$. Denote the resulting random tournament by T_p. Thus,

10970118, 2025, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/jgt.23257 by Royal Danish Library, Wiley Online Library on [12/09/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

$$\begin{aligned}\mathbb{E}(\mathrm{ism}(D, T_p)) &= \sum_{u\in V(D)} \mathbb{E}\left(\mathrm{ism}_{u\to v^*}(D, T_p)\right)\\ &= \sum_{u\in V(D)} (|D|-1)!2^{-\|D\|+d^-(u)+d^+(u)}\\ &\quad (1/2+p)^{d^+(u)}(1/2-p)^{d^-(u)}\\ &= (|D|-1)!2^{-\|D\|}\sum_{u\in V(D)}(1+2p)^{d^+(u)}\\ &\quad (1-2p)^{d^-(u)}.\end{aligned} \tag{1}$$

Observe that $\mathrm{aut}(D) = \mathrm{aut}(-D)$. Thus, for any $p \in [-1/2, 1/2]$ and $T^* \in T_p$, we have $\mathrm{ism}(D, T^*) = \mathrm{ism}(-D, T^*)$ because $f_{T^*}(D) = f_{T^*}(-D)$ and $f_{T^*}(D) = \frac{\mathrm{ism}(D,T^*)}{\mathrm{aut}(D)}$. Therefore, $\mathbb{E}(\mathrm{ism}(D, T_p)) = \mathbb{E}(\mathrm{ism}(-D, T_p))$ for any $p \in [-1/2, 1/2]$.

Then by Equation (1), for any $p \in [-1/2, 1/2]$ we have

$$\begin{aligned}&\sum_{u\in V(D)}(1+2p)^{d^+(u)}(1-2p)^{d^-(u)}\\ &\quad = \sum_{u\in V(D)}(1+2p)^{d^-(u)}(1-2p)^{d^+(u)}.\end{aligned} \tag{2}$$

As Equation (2) holds for infinitely many p, $P_D(x) = P_{-D}(x)$. □

The following corollary can be obtained from Theorem 1.3 by taking $x = 1$.

Corollary 2.1. *Let D be an orgraph, S_1 the set of its sources, and S_2 the set of its sinks. If D is converse invariant, then*

$$\sum_{v\in S_1} 2^{d^+(v)} = \sum_{v\in S_2} 2^{d^-(v)}.$$

A polynomial $f(x)$ is *even* if $f(x) = f(-x)$ for all x in the domain of f. Let $c_t(D)$ denote the coefficient of term x^t in $P_D(x)$. Then

$$c_t(D) = \sum_{v\in V(D)}\sum_{i=0}^{t}\binom{d^+(v)}{i}\binom{d^-(v)}{t-i}(-1)^{t-i}.$$

Observe that $P_D(x) = P_{-D}(x)$ if and only if the polynomial $P_D(x)$ is even. Since $P_D(x)$ is even if and only if the coefficient $c_t(D)$ of term x^t is zero for any odd number $t \le \Delta(D)$, we have the following remark.

Remark 2.2. Let D be an orgraph. Then, $P_D(x) = P_{-D}(x)$ if and only if $c_t(D) = 0$ for any odd number $t \le \Delta(D)$.

By Theorem 1.3 and Remark 2.2, we have the following corollary.

Corollary 2.3. *Let D be a converse invariant orgraph. If $\Delta(D)$ is even, then*

$$\begin{aligned}c_{\Delta(D)-1}(D) &= \sum_{v:d(v)=\Delta(D)}\left((-1)^{d^-(v)}d^+(v) + (-1)^{d^-(v)-1}d^-(v)\right)\\ &\quad + \sum_{v:d(v)=\Delta(D)-1}(-1)^{d^-(v)} = 0.\end{aligned} \tag{3}$$

If $\Delta(D)$ is odd, then

$$c_{\Delta(D)}(D) = \sum_{v:d(v)=\Delta(D)}(-1)^{d^-(v)} = 0. \tag{4}$$

By Equation (4), the following holds.

Corollary 2.4. *Let G be a graph with $\Delta(G) \ge 3$. If $\Delta(G)$ is odd and G has an odd number of vertices with the maximum degree, then for any oriented D of G, there exists a tournament T such that $f_T(D) \ne f_T(-D)$.*

We can also easily show the following result about the orientations of regular graphs by using Theorem 1.3.

Theorem 2.5. *Let d be a positive integer and D an orientation of a d-regular graph. If $\mathrm{Deg}(D) \ne \mathrm{Deg}(-D)$, then there exists a tournament T such that $f_T(D) \ne f_T(-D)$.*

Proof. Suppose to the contrary that $f_T(D) = f_T(-D)$ for every tournament T. Then by Theorem 1.3, when $x \ne 1$ we have

$$\begin{aligned}\sum_{u\in V(D)}\left(\frac{1+x}{1-x}\right)^{d^+(u)} &= \frac{P_D(x)}{(1-x)^{d(u)}} = \frac{P_{-D}(x)}{(1-x)^{d(u)}}\\ &= \sum_{u\in V(D)}\left(\frac{1+x}{1-x}\right)^{d^-(u)},\end{aligned}$$

which implies the two polynomials $\sum_{u\in V(D)}y^{d^+(u)}$ and $\sum_{u\in V(D)}y^{d^-(u)}$ are the same. Hence, $\mathrm{Deg}^+(D) = \mathrm{Deg}^-(D)$. Let $a^D_{i,d-i}$ denote the number of vertices with out-degree i and in-degree $d-i$ in D. Note that $a^D_{i,d-i} = |\{v : d^+(v) = i\}|$ and $a^{-D}_{i,d-i} = a^D_{d-i,i} = |\{v : d^-(v) = i\}|$. As $\mathrm{Deg}^+(D) = \mathrm{Deg}^-(D)$, we have that $a^D_{i,d-i} = a^{-D}_{i,d-i}$ for all $i \in [d]$ and so $\mathrm{Deg}(D) = \mathrm{Deg}(-D)$, a contradiction. □

To end this section, we now give the proof of Theorem 1.4.

Proof of Theorem 1.4. We first assume that G is a tree. Then we orient it from any leaf u such that apart from u, every other vertex has in-degree exactly one. The resulting orgraph is denoted by D. Then, we have one source u with out-degree one and at least two sinks with in-degree one in D since $\Delta(G) \ge 3$. Let S_1 be the set of sources in D and S_2 the set of its sinks. As

$$\sum_{v\in S_1} 2^{d^+(v)} = 2 < 2 + 2 \le \sum_{v\in S_2} 2^{d^-(v)},$$

we are done by applying Corollary 2.1.

We now assume that G contains cycles. Let C be one of the shortest cycles in G. Assume without loss of generality that $V(G) = \{v_1, v_2, ..., v_n\}$ and $C = v_1v_2...v_gv_1$. Note that $g < n$ since $\Delta(G) \ge 3$ and C is a shortest cycle. We obtain an orientation D of G by setting $(v_g, v_1) \in A(D)$ and $(v_i, v_j) \in A(D)$ if $i < j$ and $v_iv_j \in E(G)\backslash\{v_1v_g\}$. Let T be a tournament obtained from a strong tournament T_0 with $|T_0| = g$ and a transitive tournament T_1 with $|T_1| = n - g$ by adding all arcs from T_0 to T_1. One can observe that $f_T(D) > 0$ but $f_T(-D) = 0$, which completes the proof. □

10970118, 2025, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/jgt.23257 by Royal Danish Library, Wiley Online Library on [12/09/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

3 | Trees With Diameter at Most Three

In this section, we characterize all orientations of trees with diameter at most three that are converse invariant.

Before showing the characterization, we introduce a way of constructing converse invariant orgraphs. It is clear that if $D \cong -D$, then D is converse invariant. The following operation, first introduced in [15], provides us with a less trivial condition. For every orgraph D and $u \in V(D)$, we can obtain a new orgraph $2D_u^+$ by taking two copies of D and then adding an arc between the two vertices corresponding to u. And we say that $2D_u^+$ is obtained by *bridge-mirroring* D at vertex u. It is not hard to show that if D is converse invariant, then $2D_u^+$ is also converse invariant for every $u \in V(D)$. In fact, one can show that this statement holds for arbitrary orgraphs obtained from D by adding an arc between two nonadjacent symmetric vertices, where we say that u and v are *symmetric* if D has an automorphism ϕ such that $\phi(u) = v$ and $\phi(v) = u$.

In [14], El Sahili and Ghazo Hanna showed that $\overrightarrow{K}^d_{1,d}$ with $d \geq 3$ is not converse invariant by constructing a special tournament. The following corollary generalizes this result avoiding an explicit construction.

Corollary 3.1. *Let D be an orientation of star $K_{1,d}$ with $d \geq 3$. Then D is converse invariant if and only if $D \cong -D$.*

Proof. Let v be the central vertex of $K_{1,d}$. If D is converse invariant, then by Corollary 2.1, $d^+(v) = d^-(v)$ and therefore $D \cong -D$ when v is not a sink or source. And v cannot be a source or sink since otherwise, by Corollary 2.1, we have $2^d = 2d$ which contradicts the fact that $d \geq 3$. □

We now characterize all orientations of trees with diameter 3 that are converse invariant. We also call a tree with diameter 3 a *double star* and the *central vertices* of a double star are the vertices with degree at least 2.

Theorem 3.2. *Let D be an orientation of a double star that is not a path. Then D is converse invariant if and only if $D \cong -D$ or D is formed by bridge-mirroring $\overrightarrow{K}^0_{1,2}$ or $\overrightarrow{K}^2_{1,2}$ at the central vertex.*

Proof. Let u and v be the central vertices of D. Since D is not an orientation of a path, we have $d(u) \geq 2, d(v) \geq 2$ and $\max\{d(u), d(v)\} \geq 3$. Without loss of generality, we assume $(u, v) \in A(D)$. Thus, u cannot be a sink and v cannot be a source. Suppose that D is converse invariant. We consider the following three cases.

Case 1. Exactly one of u and v is a sink or source.

Note that there are two subcases: (i) u is a source and v is not a sink, and (ii) u is not a source and v is a sink. Now we consider Subcase (i). By Corollary 2.1, we have $2^{d^+(u)} + 2(d^-(v) - 1) = 2(d^+(u) - 1) + 2d^+(v)$ and so

$$2^{d^+(u)} - 2d^+(u) = 2(d^+(v) - d^-(v)). \tag{5}$$

Suppose that $d^+(u) = 2$. Then $d^+(v) = d^-(v)$ by Equation (5). Since $\max\{d(u), d(v)\} \geq 3$, we have $d(v) \geq 4$. Assume that $d^+(v) = d^-(v) = n - 1$. Then, $|V(D)| = 2n$. Note that $n \geq 3$ as $d(v) \geq 4$. Let T be a tournament with $V(T) = \{v_1, v_2, ..., v_{2n}\}$ and $A(T) = (\{(v_i, v_j) : 1 \leq i < j \leq 2n\} \backslash \{(v_1, v_n)\}) \cup \{(v_n, v_1)\}$. Now we consider $f_T(D)$ and $f_T(-D)$. Note that in T, only the out-degree and in-degree of v_{n+1} are both at least $n - 1$. Then the vertex v of D can only be mapped to the vertex v_{n+1} of T. A simple calculation shows that $f_T(D) = \frac{n(n-1)}{2}$ and $f_T(-D) = n(n - 1)$. Therefore, $f_T(D) \neq f_T(-D)$, a contraction.

Suppose that $d^+(u) \geq 3$. Thus, if $d^-(v) \geq 2$, then by Equation (5) we have

$$\begin{aligned} d^+(v) + d^-(v) - 1 \geq d^+(v) + 1 \geq 2^{d^+(u)-1} - d^+(u) \\ + 3 \geq d^+(u) + 1, \end{aligned} \tag{6}$$

which implies $d(v) = d^+(v) + d^-(v) \geq d^+(u) + 2 = d(u) + 2$. Then by Equation (4), $d(v)$ is even. And now, by Equation (3) and the fact that $d(v) \geq d(u) + 2$ (and therefore $\{t : d(t) = \Delta(D) - 1\} = \emptyset$), we have $d^+(v) = d^-(v)$ which contradicts Equation (5) and the fact that $d^+(u) \geq 3$. Note that if $d^-(v) = 1$ and $d^+(v) \geq d^+(u) + 1$, then $d(v) \geq d(u) + 2$ also holds and we can use the same argument.

It remains to consider the case when $d^-(v) = 1$ and $d^+(v) \leq d^+(u)$. By (5), we have $d^+(v) = 2^{d^+(u)-1} - d^+(u) + 1$. Since $d^+(v) \leq d^+(u)$, we have $d^+(u) = 3$ and therefore $d^+(v) = 2$. This implies that D is obtained by bridge-mirroring $\overrightarrow{K}^2_{1,2}$ at the central vertex. This completes Subcase (i).

Observe that Subcase (ii) is Subcase (i) for $-D$ instead of D. Thus, in Subcase (ii) we can show that D is obtained by bridge-mirroring $\overrightarrow{K}^0_{1,2}$ at the central vertex.

Case 2. u is a source and v is a sink.

By Corollary 2.1, we have $2^{d^+(u)} + 2d^-(v) - 2 = 2^{d^-(v)} + 2d^+(u) - 2$ and therefore

$$2^{d^+(u)} - 2d^+(u) = 2^{d^-(v)} - 2d^-(v).$$

This implies $d^+(u) = d^-(v)$ and thus $D \cong -D$.

Case 3. u is not a source and v is not a sink.

By Corollary 2.1, we have that the number of sinks equal to the number of sources which means

$$d^+(u) + d^+(v) = d^-(u) + d^-(v). \tag{7}$$

Suppose that $d^+(v) = d^-(u)$. Then $d^+(u) = d^-(v)$ by Equation (7) and therefore $D \cong -D$. Suppose that $d^+(v) \neq d^-(u)$. Assume that $d^+(v) > d^-(u)$ (the case $d^+(v) < d^-(u)$ can be proved similarly). Since u is not a source, $d^-(u) \geq 1$. Let T be a transitive tournament with $V(T) = \{v_1, v_2, ..., v_{|D|}\}$ and $A(T) = \{(v_i, v_j) : 1 \leq i < j \leq |D|\}$. Let T' be a tournament with $V(T') = V(T)$ and $A(T') = A(T) \backslash \{(v_1, v_{d^-(u)+1})\}) \cup \{(v_{d^-(u)+1}, v_1)\}$.

10970118, 2025, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/jgt.23257 by Royal Danish Library, Wiley Online Library on [12/09/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Note that $T' \ncong T$ because $d^-(u) + 1 \geq 2$. We first consider $f_{T'}(D)$. Since $(u, v) \in A(D)$, u is mapped to a vertex in T' with a subscript less than the subscript of the vertex to which v is mapped in T' for any copy of D in T'. Let a denote the number of copies of D in T where the vertex u in D is mapped to the vertex $v_{d^-(u)+1}$ in T. Note that $a \geq 1$. Then $f_{T'}(D) = f_T(D) - a$. Since $T \cong -T$, we have $f_T(D) = f_T(-D)$ and thus $f_{T'}(D) = f_T(-D) - a$. Now we consider $f_{T'}(-D)$. As $d^+(v) > d^-(u)$, in any copy of $-D$ in T', v must have been mapped to a vertex with a higher subscript than $d^-(u) + 1$ and therefore $f_{T'}(-D) = f_T(-D)$. Then, $f_{T'}(D) = f_T(-D) - a = f_{T'}(-D) - a$. Hence, $f_{T'}(D) \neq f_{T'}(-D)$, a contradiction. □

Corollary 3.1 and Theorem 3.2 imply Theorem 1.5.

4 | Conclusion

In this paper, we introduce a new digraph polynomial and use it to show a necessary condition for an orgraph to be converse invariant. In particular, we use this necessary condition to characterize converse invariant orientations of trees with diameters bounded by three. We would like to make the following conjecture.

Conjecture 4.1. *Let D be an orientation of a tree with maximum degree at least 3. Then D is converse invariant if and only if $D \cong -D$ or D can be obtained by the bridge-mirroring operation recursively from an orientation of a path.*

An *out-branching* B_s^+ (respectively, *in-branching* B_s^-) is an orientation of a tree in which each vertex $x \neq s$ has precisely one arc entering (leaving) it and s has no arcs entering (leaving) it. The vertex s is the root of B_s^+ (respectively, B_s^-). Conjecture 4.1 might be difficult to verify. As a special case, one can consider out-branchings or in-branchings for Conjecture 4.1.

Acknowledgments

Jiangdong Ai: Supported by National Natural Science Foundation of China under grant No. 12161141006 and No. 12401456. Hui Lei: Partially supported by the NSFC grant (No. 12371351).

Data Availability Statement

The authors have nothing to report.

References

1. J. Bang-Jensen and G. Gutin, *Digraphs: Theory, Algorithms and Applications* (Springer-Verlag, 2009). 2nd ed.

2. T. Szele, "Kombinatorikai vizsgalatok az iranyiott tejles graffal kapcsolatban," *Acta Scientiarum Mathematicarum* 50 (1943): 223–256.

3. N. Alon, "The Maximum Number of Hamiltonian Paths in Tournaments," *Combinatorica* 10, no. 4 (1990): 319–324.

4. I. Adler, N. Alon, and S. M. Ross, "On the Maximum Number of Hamiltonian Paths in Tournaments," *Random Structures & Algorithms* 18, no. 3 (2001): 291–296.

5. E. Friedgut and J. Kahn, "On the Number of Hamiltonian Cycles in a Tournament," *Combinatorics, Probability and Computing* 14, no. 5–6 (2005): 769–781.

6. J. W. Moon, "The Minimum Number of Spanning Paths in a Strong Tournament," *Publicationes Mathematicae Debrecen* 19 (1972): 101–104.

7. A. H. Busch, "A Note on the Number of Hamiltonian Paths in Strong Tournaments," *Electronic Journal of Combinatorics* 13 (2006): N3.

8. B. Cuckler, "Hamiltonian Cycles in Regular Tournaments," *Combinatorics, Probability and Computing* 16, no. 2 (2007): 239–249.

9. A. Ferber and E. Long, "Packing and Counting Arbitrary Hamilton Cycles in Random Digraphs," *Random Structures & Algorithms* 54, no. 3 (2019): 499–514.

10. A. Ferber, M. Kwan, and B. Sudakov, "Counting Hamilton Cycles in Sparse Random Directed Graphs," *Random Structures & Algorithms* 53, no. 4 (2018): 592–603.

11. C. Thomassen, "On the Number of Hamiltonian Cycles in Tournaments," *Discrete mathematics* 31, no. 3 (1980): 315–323.

12. R. Yuster, "On the Maximum Number of Spanning Copies of an Orientation in a Tournament," *Combinatorics, Probability and Computing* 26, no. 5 (2017): 775–796.

13. M. Rosenfeld, "Antidirected Hamiltonian Circuits in Tournaments," *Journal of Combinatorial Theory, Series B* 16, no. 3 (1974): 234–242.

14. A. El Sahili and Z. Ghazo Hanna, "About the Number of Oriented Hamiltonian Paths and Cycles in Tournaments," *Journal of Graph Theory* 102 (2023): 684–701.

15. Y. Zhao and Y. Zhou, "Impartial Digraphs," *Combinatorica* 40, no. 6 (2020): 875–896.

10970118, 2025, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/jgt.23257 by Royal Danish Library, Wiley Online Library on [12/09/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License